
BY 516: Introduction to “OMICS” and Systems Analysis

Course Code	BY 516
Course Name	Introduction to “OMICS” and Systems Analysis
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to OMICS**

- i. Genomics
 - a. High throughput Next-generation sequencing methods
 - b. Quality filtering and reads assembly
- ii. Epigenomics
 - a. Bisulfite sequencing
 - b. ChIP assays
- iii. Transcriptomics
 - a. RNA-seq
- iv. Metagenomics and metatranscriptomics
 - a. Phylo-typing
 - b. Binning [12 Lectures]

- **Methods in Systems Analysis**

Basic concepts in systems biology, introduction to Gene Ontology, KEGG and EcoCyc database; Reconstruction of biochemical networks including metabolic networks, transcriptional regulatory networks, and signaling networks; Genome scale modelling, from networks to constraint based models, mathematical representation of reconstructed networks, basic features of Stoichiometric matrix and topological properties analysis; Application of systems analysis to Microbial organisms, analysis of Pathways, Metabolic network properties, Metabolic control analysis, Simulation of cellular activities, Gene dispensability in metabolism. [30 Lectures]

Text and Reference Books

1. Bernhard O. Palsson, **Systems Biology: Properties of Reconstructed Networks**, Cambridge University Press, 2006.
2. Victor Kunin et al., **A Bioinformatician's Guide to Metagenomics**, MICROBIOLOGY AND MOLECULAR BIOLOGY REVIEWS, Dec. 2008, p. 557–578.
3. Next-generation sequencing data interpretation: enhancing reproducibility and accessibility, **Anton Nekrutenko & James Taylor Nature Reviews Genetics**, September 2012 Volume 13, No 9, 667.
4. SOberhardt MA et al., Applications of genome-scale metabolic reconstructions, **Mol Syst Biol**, 5:320, 2009.
5. Francke C et. al., Reconstructing the metabolic network of a bacterium from its genome, **Trends Microbiol**, 13:550-8, 2005.